Removable the limit exists; the fault is a single value, so redefining $f(a)$ makes the function continuous

- **Hole** $f(a)$ is undefined — $(x^2-1)/(x-1)$ at $x=1$: limit 2, no value
- **Misplaced value** $f(a)$ exists but differs from the limit — the point sits off the curve

Non-removable the limit does not exist; no single value of $f(a)$ can fix it

- **Jump** both one-sided limits exist but differ — $\lfloor x \rfloor$ at $x=2$: 1 from the left, 2 from the right
- **Infinite** at least one one-sided limit is $\pm\infty$ — $1/x$ at $x=0$
- **Oscillating** the limit fails to exist because the function oscillates without settling — $\sin(1/x)$ at $x=0$

- [ ] the limit exists: one value repairs it
- [ ] the limit does not exist: nothing repairs it

FOUR DISCONTINUITIES, TWO GROUPS: REPAIRABLE OR NOT · /CALCULUS/LIMITS/CONTINUITY

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